CURRICULUM VITAE

October 24, 2026

David LACOSTE

born on december 22nd 1971 in Montpellier (France),
CNRS researcher in the Laboratoire Gulliver,
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EDUCATION

2009	Habilitation degree from P. et M. Curie University
1996 – 1999	Ph.D. in Theoretical Physics, Université Joseph Fourier, Grenoble
1994 – 1995	Master in Quantum Physics from Ecole Normale Supérieure, Paris

EMPLOYMENT

Jun 2016 – present	CNRS Directeur de Recherches (DR2 level) at Gulliver laboratory, ESPCI
Jun 2005 – Jun 2016	CNRS Chargé de Recherches (CR1 level) at Gulliver laboratory, ESPCI
Jan 2002 – Jun 2005	CNRS Chargé de Recherches (CR2 level) at Gulliver laboratory, ESPCI
Jan 2000 – Jan 2002	Postdoctoral Fellow with T. Lubensky at Dept. of Physics, University of Pennsylvania, Philadelphia, USA
1992 – 1995	Undergraduate studies at Ecole Normale Supérieure, Paris

HONOR

Sept 2020	Distinction "Outstanding referee" for APS
Sept 2012	Distinction "Outstanding referee" for EPL
Sept 2011 – 2015	Distinction "Prime d'excellence scientifique" from CNRS

RESEARCH

I am a theoretical physicist working on the statistical physics of living matter and on the emergence of life. My research spans stochastic thermodynamics, the origin of life, chemical reaction networks and autocatalysis, and the growth and fluctuations of cell populations. I am author of 81 peer-reviewed publications (h index 28), which are available on my web site.

SELECTED PUBLICATIONS

[1] A. Blokhuis, D. Lacoste and P. Nghe, *Universal motifs and the diversity of autocatalytic systems*, Proc. Natl. Acad. Sci. U.S.A. **117**, 25230–25236 (2020).

[2] L.-F. Rojas-Ochoa, D. Lacoste, R. Lenke, P. Schurtenberger and F. Scheffold, *Depolarization of backscattered linearly polarized light*, J. Opt. Soc. Am. A **21**, 1799–1804 (2004).

[3] F. Quemeneur, J. K. Sigurdsson, M. Renner, P. J. Atzberger, P. Bassereau and D. Lacoste, *Shape matters in protein mobility within membranes*, Proc. Natl. Acad. Sci. U.S.A. **111**, 5083–5087 (2014).

[4] A. W. C. Lau, D. Lacoste and K. Mallick, *Nonequilibrium fluctuations and mechanochemical couplings of a molecular motor*, Phys. Rev. Lett. **99**, 158102 (2007).

[5] M. F. Islam, K. H. Lin, D. Lacoste, T. C. Lubensky and A. G. Yodh, *Field-induced structures in miscible ferrofluid suspensions with and without latex spheres*, Phys. Rev. E **67**, 021402 (2003).

STUDENTS SUPERVISION

Currently supervising 1 PhD student, B. Ledoux. A. Despons recently defended his thesis on November 21st on the topic of 'Physical aspects of autocatalysis'. I have supervised so far 6 PhD students and about 9 postdocs.

ACTIVITIES OF GENERAL INTEREST

- Member of the organizing committee of Qlife, a funding initiative of PSL University in Biophysics.
- Referee for PRL, PRE, EuroPhys. Letters, New J. of Phys., J. Chem. Phys., J. Stat. Mech...
- Organization of international conferences 3 between 2017 and 2018 and 2 in 2024 including 'Sadi Carnot's legacy' at Ecole Polytechnique on September 16th-18th 2024.

LIST OF INVITED TALKS IN THE LAST TEN YEARS

- Evolution in Paris, 30 Sept 2024
- Paris-Rice meeting, Sept 2024
- Thermodynamics 2.0 (on line), Broomfield, USA, August 2024
- Workshop on Emergence of Information in Molecular Systems, Munich, July 2024
- Workshop on Evolution of population dynamics, Pasteur Institute, Paris, June 2024
- Congress of French Society of Physics, Paris, July 2023
- Postmodern Thermodynamics conference, Luxembourg, December 2022
- Thermodynamics 2.0 (on line), Riverside USA, July 2022
- Stochastic Thermodynamics conference (on line), ICTP, Bangalore, June 2022
- Physics of the Emergence of Life, MIAPP program, Munich September 2021
- Joint European Thermodynamics Conference (JETC), Prag, June 2021
- Conference of the Mathematical Biology Society (SMS), Riverside, USA, June 2021
- Conference of French physics society (SFP), Nantes, July 2019
- Workshop on "Random talks in Stochastic Thermodynamics", Santa Marinella, June 2019
- Workshop on the origin of biochemistry, Florence, Italy, October 2018
- Conference in honor of L. Peliti "A path in statistical physics", Univ. Paris Diderot, September 2018
- Conference on Stochastic Thermodynamics, Dresden, September 2018
- Conference on Stochastic Thermodynamics, Stockholm, Septembre 2017
- Conference on Evolution and Diversity, NCBS Bangalore, India, February 2017
- Non-equilibrium and non-linear phenomena in Statistical Mechanics, Brussels, July 2016
- Biomembrane workshop, UC Davis, USA, September 2016
- Statistical Physics Conference, Lyon, July 2016
- Joint European Thermodynamics Conference (JETC), Nancy, May 2015
- Workshop "Stochastic Thermodynamics of biological systems", Stockholm, September 2015
- "Fluctuation theorems: where do we go from here",
- Workshop "Thermodynamics of out-of-equilibrium quantum systems", Grenoble October 2014
- Conference Condensed Matter CMD25, Paris, August 2014.